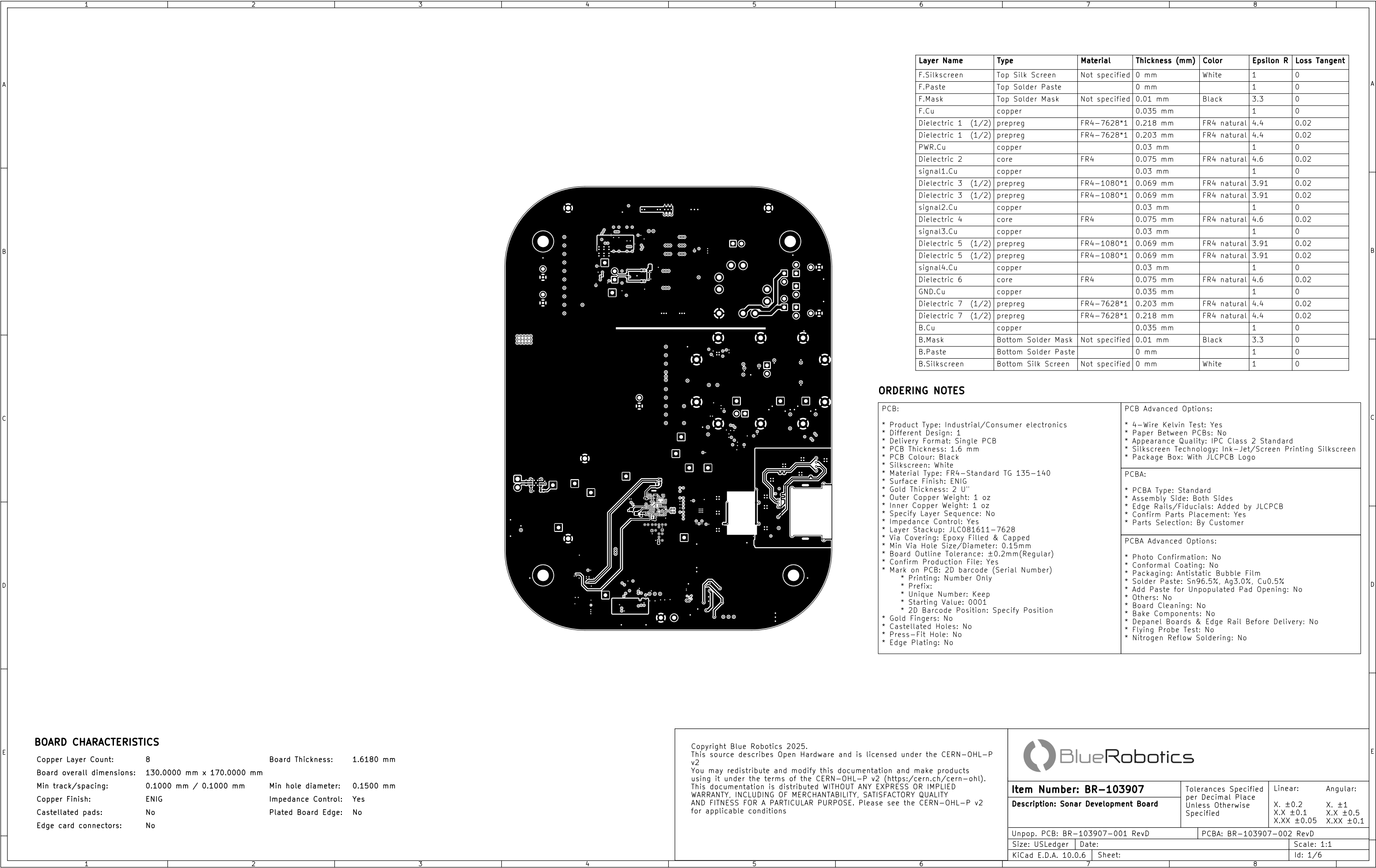
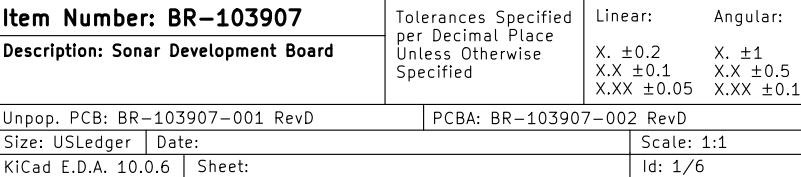


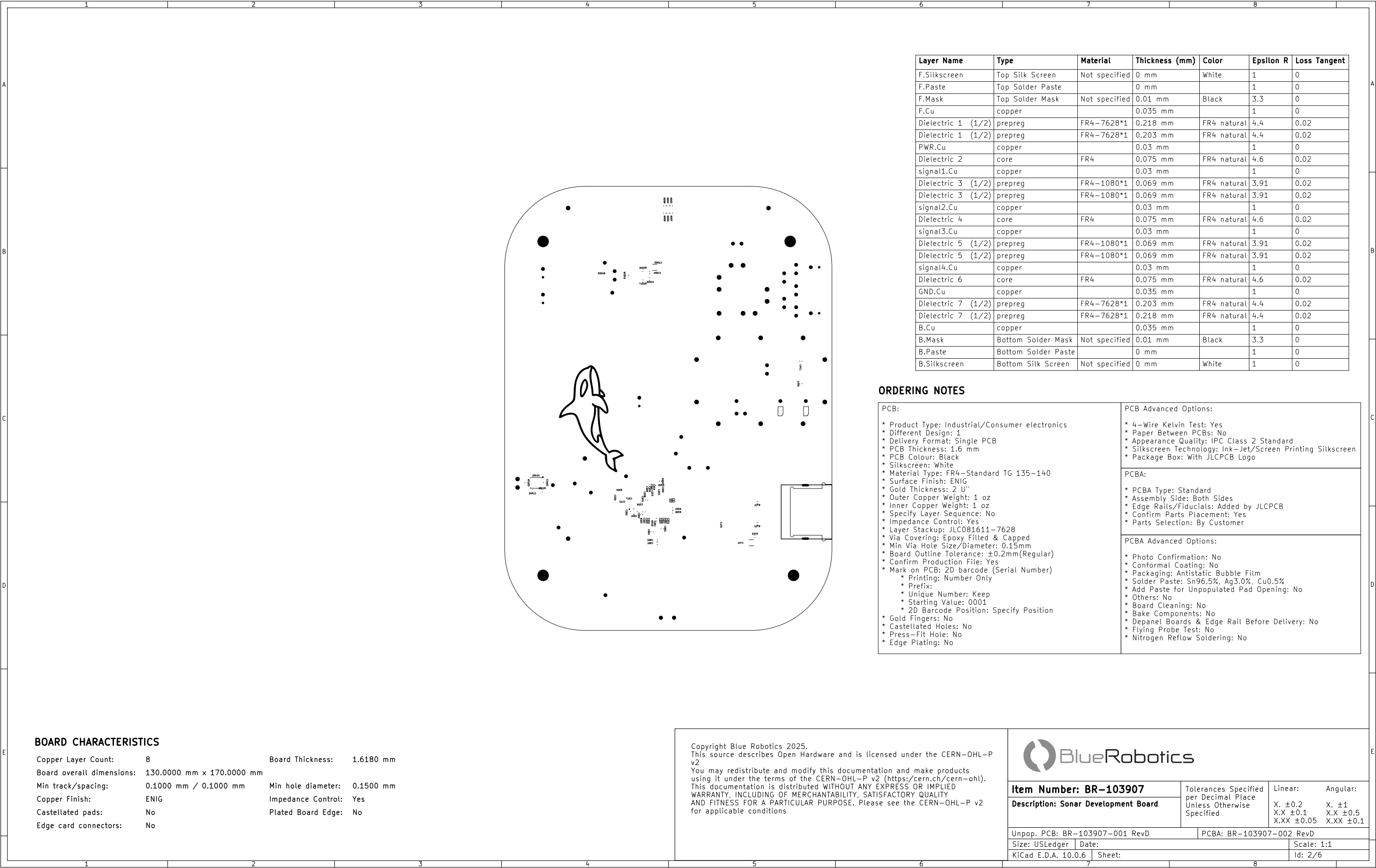


ORDERING NOTES

BOARD CHARACTERISTICS

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Layer Name	Type	Material	Thickness (mm)	Color	Epsilon R	Loss Tangent
F.Silkscreen	Top Silk Screen	Not specified	0 mm	White	1	0
F.Paste	Top Solder Paste		0 mm		1	0
F.Mask	Top Solder Mask	Not specified	0.01 mm	Black	3.3	0
F.Cu	copper		0.035 mm		1	0
Dielectric 1 (1/2)	prepreg	FR4-7628*1	0.218 mm	FR4 natural	4.4	0.02
Dielectric 1 (1/2)	prepreg	FR4-7628*1	0.203 mm	FR4 natural	4.4	0.02
PWR.Cu	copper		0.03 mm		1	0
Dielectric 2	core	FR4	0.075 mm	FR4 natural	4.6	0.02
signal1.Cu	copper		0.03 mm		1	0
Dielectric 3 (1/2)	prepreg	FR4-1080*1	0.069 mm	FR4 natural	3.91	0.02
Dielectric 3 (1/2)	prepreg	FR4-1080*1	0.069 mm	FR4 natural	3.91	0.02
signal2.Cu	copper		0.03 mm		1	0
Dielectric 4	core	FR4	0.075 mm	FR4 natural	4.6	0.02
signal3.Cu	copper		0.03 mm		1	0
Dielectric 5 (1/2)	prepreg	FR4-1080*1	0.069 mm	FR4 natural	3.91	0.02
Dielectric 5 (1/2)	prepreg	FR4-1080*1	0.069 mm	FR4 natural	3.91	0.02
signal4.Cu	copper		0.03 mm		1	0
Dielectric 6	core	FR4	0.075 mm	FR4 natural	4.6	0.02
GND.Cu	copper		0.035 mm		1	0
Dielectric 7 (1/2)	prepreg	FR4-7628*1	0.203 mm	FR4 natural	4.4	0.02
Dielectric 7 (1/2)	prepreg	FR4-7628*1	0.218 mm	FR4 natural	4.4	0.02
B.Cu	copper		0.035 mm		1	0
B.Mask	Bottom Solder Mask	Not specified	0.01 mm	Black	3.3	0
B.Paste	Bottom Solder Paste		0 mm		1	0
B.Silkscreen	Bottom Silk Screen	Not specified	0 mm	White	1	0

ORDERING NOTES

PCB: * Product Type: Industrial/Consumer electronics * Different Design: 1 * Delivery Format: Single PCB * PCB Thickness: 1.6 mm * PCB Colour: Black * Silkscreen: White * Material Type: FR4—Standard TG 135—140 * Surface Finish: ENIG * Gold Thickness: 2 U" * Outer Copper Weight: 1 oz * Inner Copper Weight: 1 oz * Specify Layer Sequence: No * Impedance Control: Yes * Layer Stackup: JLC081611—7628 * Via Covering: Epoxy Filled & Capped * Min Via Hole Size/Diameter: 0.15mm * Board Outline Tolerance: $\pm 0.2\text{mm}$(Regular) * Confirm Production File: Yes * Mark on PCB: 2D barcode (Serial Number) * Printing: Number Only * Prefix: * Unique Number: Keep * Starting Value: 0001 * 2D Barcode Position: Specify Position * Gold Fingers: No * Castellated Holes: No * Press—Fit Hole: No * Edge Plating: No	PCB Advanced Options: * 4—Wire Kelvin Test: Yes * Paper Between PCBs: No * Appearance Quality: IPC Class 2 Standard * Silkscreen Technology: Ink—Jet/Screen Printing Silkscreen * Package Box: With JLCPCB Logo PCBA: * PCBA Type: Standard * Assembly Side: Both Sides * Edge Rails/Fiducials: Added by JLCPCB * Confirm Parts Placement: Yes * Parts Selection: By Customer PCBA Advanced Options: * Photo Confirmation: No * Conformal Coating: No * Packaging: Antistatic Bubble Film * Solder Paste: Sn96.5%, Ag3.0%, Cu0.5% * Add Paste for Unpopulated Pad Opening: No * Others: No * Board Cleaning: No * Bake Components: No * Depanel Boards & Edge Rail Before Delivery: No * Flying Probe Test: No * Nitrogen Reflow Soldering: No
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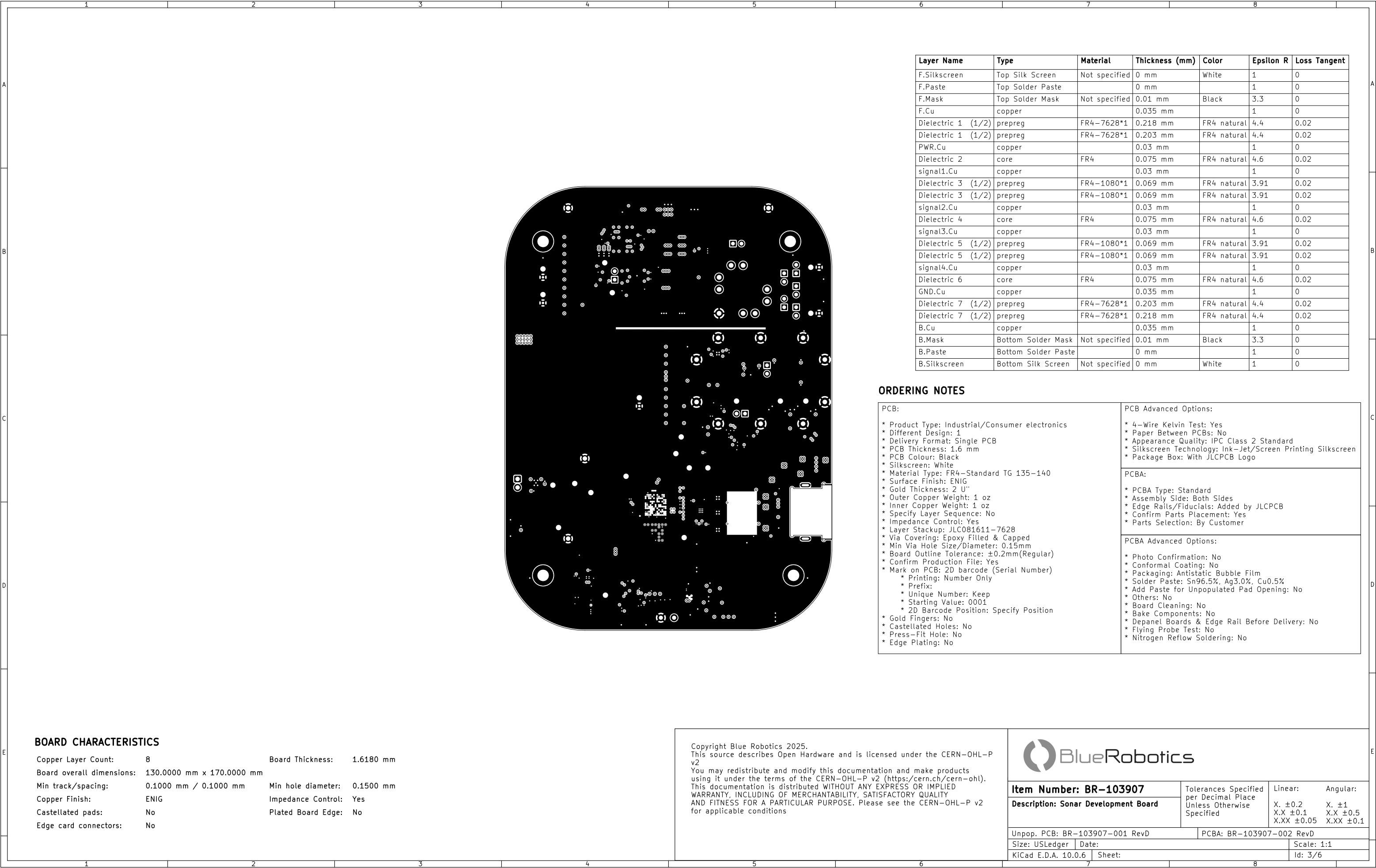
BOARD CHARACTERISTICS

Copper Layer Count:	8	Board Thickness:	1.6180 mm
Board overall dimensions:	130.0000 mm x 170.0000 mm		
Min track/spacing:	0.1000 mm / 0.1000 mm	Min hole diameter:	0.1500 mm
Copper Finish:	ENIG	Impedance Control:	Yes
Castellated pads:	No	Plated Board Edge:	No
Edge card connectors:	No		

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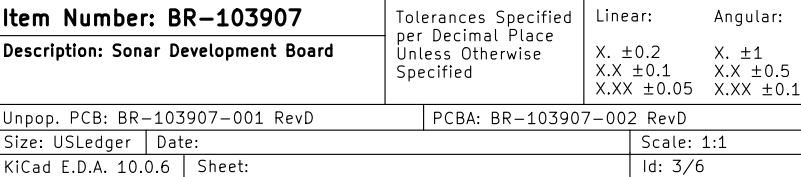
Item Number: BR-103907		Tolerances Specified per Decimal Place Unless Otherwise Specified	Linear:	Angular:
Description: Sonar Development Board			X. ± 0.2 X.X ± 0.1 X.XX ± 0.05	X. ± 1 X.X ± 0.5 X.XX ± 0.1
Unpop. PCB: BR-103907-001 RevD		PCBA: BR-103907-002 RevD		
Size: USL	Ledger	Date:	Scale: 1:1	
KiCad	E.D.A.	10.0.6	Sheet:	Id: 2/6

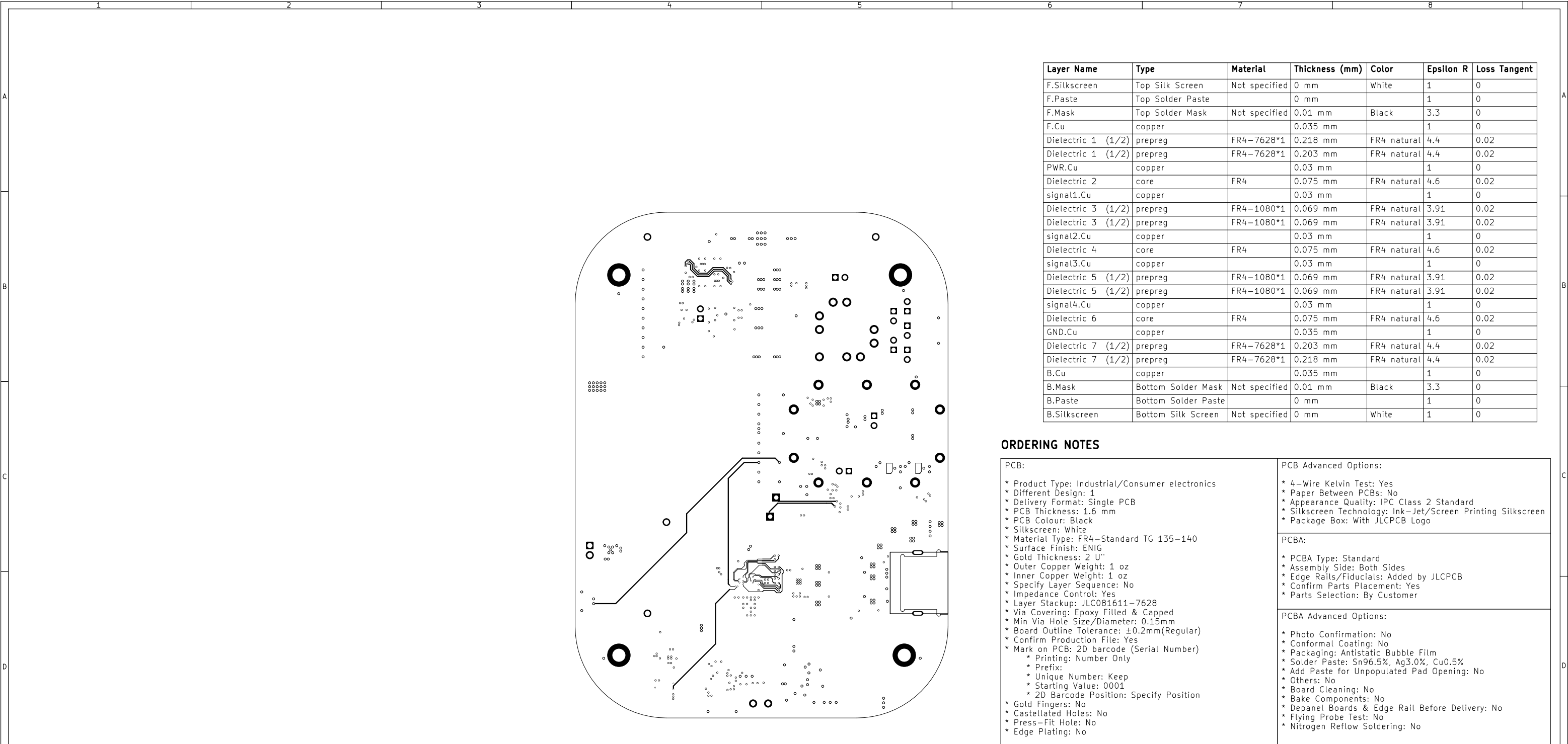


ORDERING NOTES

BOARD CHARACTERISTICS

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ORDERING NOTES

PCB: * Product Type: Industrial/Consumer electronics * Different Design: 1 * Delivery Format: Single PCB * PCB Thickness: 1.6 mm * PCB Colour: Black * Silkscreen: White * Material Type: FR4—Standard TG 135—140 * Surface Finish: ENIG * Gold Thickness: 2 U" * Outer Copper Weight: 1 oz * Inner Copper Weight: 1 oz * Specify Layer Sequence: No * Impedance Control: Yes * Layer Stackup: JLC081611—7628 * Via Covering: Epoxy Filled & Capped * Min Via Hole Size/Diameter: 0.15mm * Board Outline Tolerance: $\pm 0.2\text{mm}$(Regular) * Confirm Production File: Yes * Mark on PCB: 2D barcode (Serial Number) * Printing: Number Only * Prefix: * Unique Number: Keep * Starting Value: 0001 * 2D Barcode Position: Specify Position * Gold Fingers: No * Castellated Holes: No * Press—Fit Hole: No * Edge Plating: No	PCB Advanced Options: * 4—Wire Kelvin Test: Yes * Paper Between PCBs: No * Appearance Quality: IPC Class 2 Standard * Silkscreen Technology: Ink—Jet/Screen Printing Silkscreen * Package Box: With JLCPCB Logo PCBA: * PCBA Type: Standard * Assembly Side: Both Sides * Edge Rails/Fiducials: Added by JLCPCB * Confirm Parts Placement: Yes * Parts Selection: By Customer PCBA Advanced Options: * Photo Confirmation: No * Conformal Coating: No * Packaging: Antistatic Bubble Film * Solder Paste: Sn96.5%, Ag3.0%, Cu0.5% * Add Paste for Unpopulated Pad Opening: No * Others: No * Board Cleaning: No * Bake Components: No * Depanel Boards & Edge Rail Before Delivery: No * Flying Probe Test: No * Nitrogen Reflow Soldering: No
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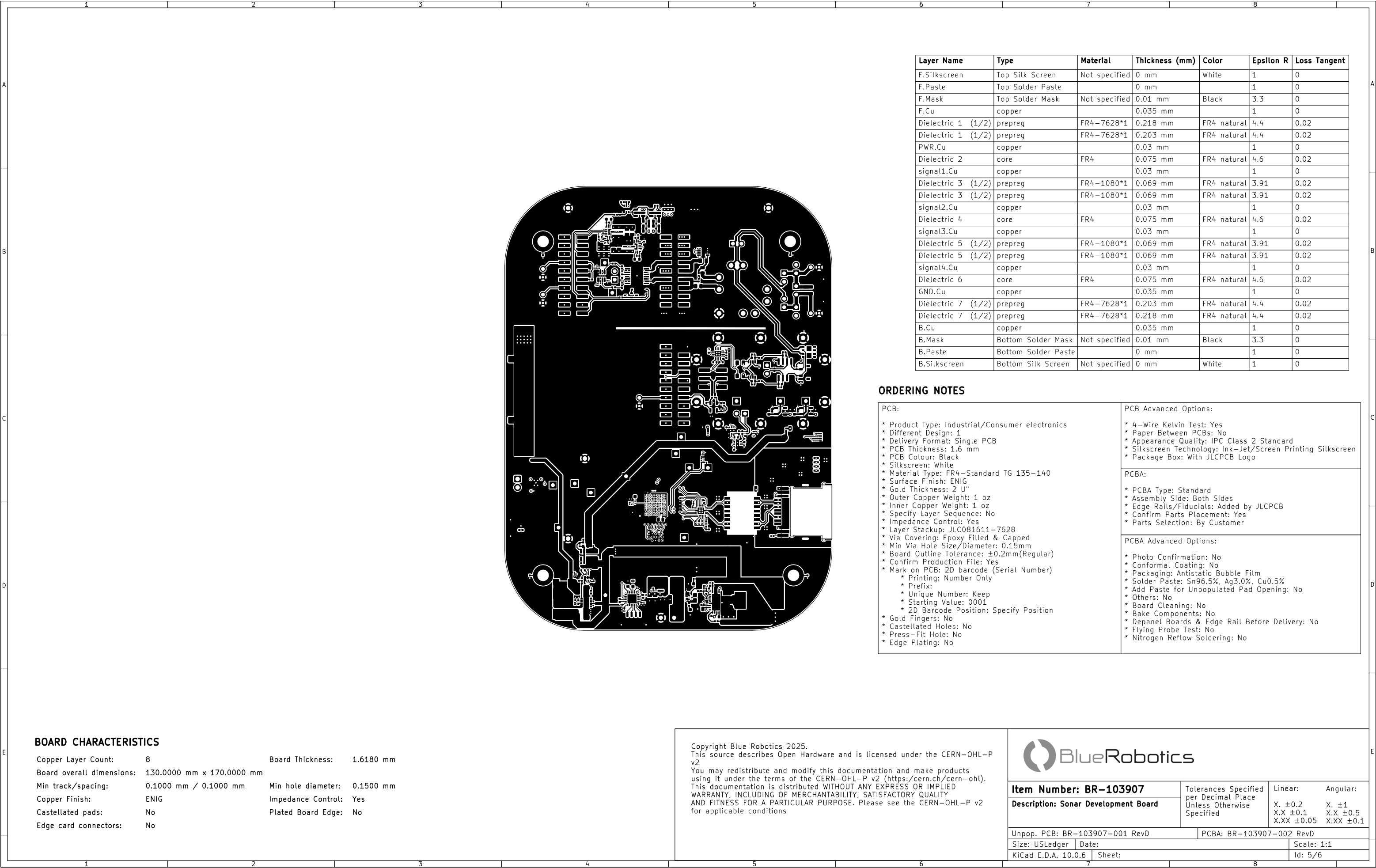
BOARD CHARACTERISTICS

Copper Layer Count:	8	Board Thickness:	1.6180 mm
Board overall dimensions:	130.0000 mm x 170.0000 mm		
Min track/spacing:	0.1000 mm / 0.1000 mm	Min hole diameter:	0.1500 mm
Copper Finish:	ENIG	Impedance Control:	Yes
Castellated pads:	No	Plated Board Edge:	No
Edge card connectors:	No		

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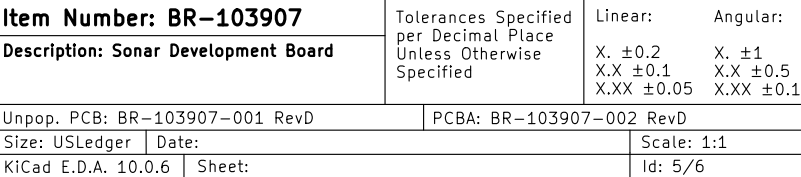
Item Number: BR-103907		Tolerances Specified per Decimal Place Unless Otherwise Specified	Linear:	Angular:
Description: Sonar Development Board			X. ± 0.2 X.X ± 0.1 X.XX ± 0.05	X. ± 1 X.X ± 0.5 X.XX ± 0.1
Unpop. PCB: BR-103907-001 RevD		PCBA: BR-103907-002 RevD		
Size: USLedge	Date:	Scale: 1:1		
KiCad E.D.A. 10.0.6	Sheet:	Id: 4/6		

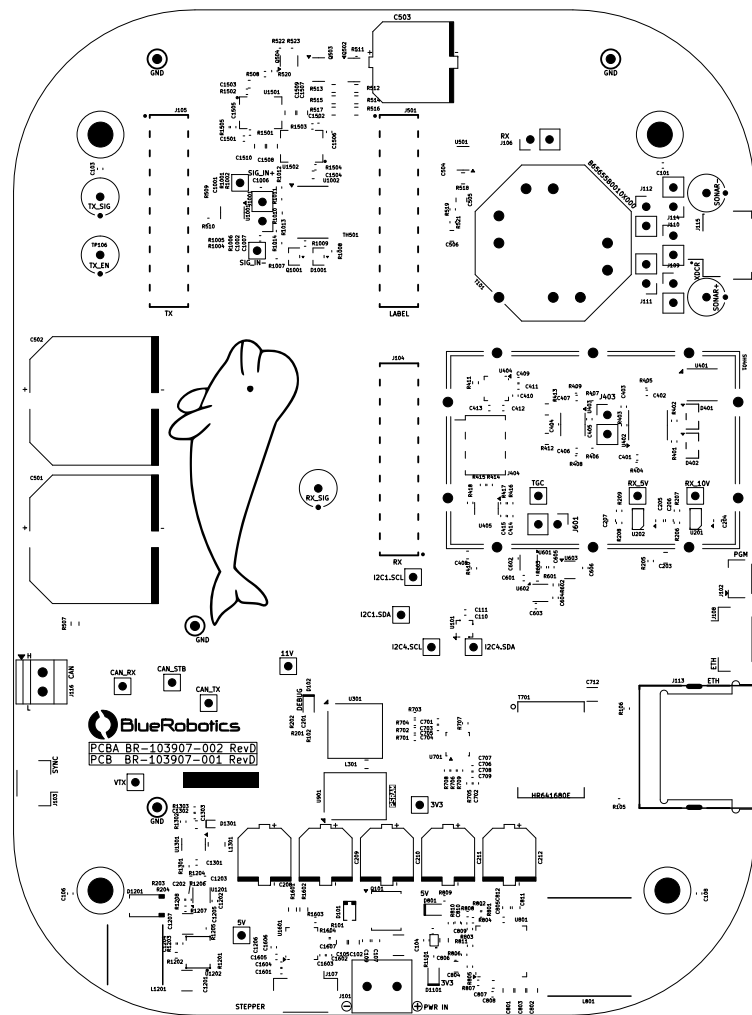


ORDERING NOTES

BOARD CHARACTERISTICS

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Layer Name	Type	Material	Thickness (mm)	Color	Epsilon R	Loss Tangent
F.Silkscreen	Top Silk Screen	Not specified	0 mm	White	1	0
F.Paste	Top Solder Paste		0 mm		1	0
F.Mask	Top Solder Mask	Not specified	0.01 mm	Black	3.3	0
F.Cu	copper		0.035 mm		1	0
Dielectric 1 (1/2)	prepreg	FR4-7628*1	0.218 mm	FR4 natural	4.4	0.02
Dielectric 1 (1/2)	prepreg	FR4-7628*1	0.203 mm	FR4 natural	4.4	0.02
PWR.Cu	copper		0.03 mm		1	0
Dielectric 2	core	FR4	0.075 mm	FR4 natural	4.6	0.02
signal1.Cu	copper		0.03 mm		1	0
Dielectric 3 (1/2)	prepreg	FR4-1080*1	0.069 mm	FR4 natural	3.91	0.02
Dielectric 3 (1/2)	prepreg	FR4-1080*1	0.069 mm	FR4 natural	3.91	0.02
signal2.Cu	copper		0.03 mm		1	0
Dielectric 4	core	FR4	0.075 mm	FR4 natural	4.6	0.02
signal3.Cu	copper		0.03 mm		1	0
Dielectric 5 (1/2)	prepreg	FR4-1080*1	0.069 mm	FR4 natural	3.91	0.02
Dielectric 5 (1/2)	prepreg	FR4-1080*1	0.069 mm	FR4 natural	3.91	0.02
signal4.Cu	copper		0.03 mm		1	0
Dielectric 6	core	FR4	0.075 mm	FR4 natural	4.6	0.02
GND.Cu	copper		0.035 mm		1	0
Dielectric 7 (1/2)	prepreg	FR4-7628*1	0.203 mm	FR4 natural	4.4	0.02
Dielectric 7 (1/2)	prepreg	FR4-7628*1	0.218 mm	FR4 natural	4.4	0.02
B.Cu	copper		0.035 mm		1	0
B.Mask	Bottom Solder Mask	Not specified	0.01 mm	Black	3.3	0
B.Paste	Bottom Solder Paste		0 mm		1	0
B.Silkscreen	Bottom Silk Screen	Not specified	0 mm	White	1	0

ORDERING NOTES

PCB:

- * Product Type: Industrial/Consumer electronics
- * Different Design: 1
- * Delivery Format: Single PCB
- * PCB Thickness: 1.6 mm
- * PCB Colour: Black
- * Silkscreen: White
- * Material Type: FR4—Standard TG 135–140
- * Surface Finish: ENIG
- * Gold Thickness: 2 U"
- * Outer Copper Weight: 1 oz
- * Inner Copper Weight: 1 oz
- * Specify Layer Sequence: No
- * Impedance Control: Yes
- * Layer Stackup: JLC081611–7628
- * Via Covering: Epoxy Filled & Capped
- * Min Via Hole Size/Diameter: 0.15mm
- * Board Outline Tolerance: $\pm 0.2\text{mm}$ (Regular)
- * Confirm Production File: Yes
- * Mark on PCB: 2D barcode (Serial Number)
 - * Printing: Number Only
 - * Prefix:
 - * Unique Number: Keep
 - * Starting Value: 0001
 - * 2D Barcode Position: Specify Position
- * Gold Fingers: No
- * Castellated Holes: No
- * Press–Fit Hole: No
- * Edge Plating: No

PCB Advanced Options:

- * 4-Wire Kelvin Test: Yes
- * Paper Between PCBs: No
- * Appearance Quality: IPC Class 2 Standard
- * Silkscreen Technology: Ink-Jet/Screen Printing Silkscreen
- * Package Box: With JLCPCB Logo

PCBA:

- * PCBA Type: Standard
- * Assembly Side: Both Sides
- * Edge Rails/Fiducials: Added by JLCPCB
- * Confirm Parts Placement: Yes
- * Parts Selection: By Customer

PCBA Advanced Options:

- * Photo Confirmation: No
- * Conformal Coating: No
- * Packaging: Antistatic Bubble Film
- * Solder Paste: Sn96.5%, Ag3.0%, Cu0.5%
- * Add Paste for Unpopulated Pad Opening: No
- * Others: No
- * Board Cleaning: No
- * Bake Components: No
- * Depanel Boards & Edge Rail Before Delivery: No
- * Flying Probe Test: No
- * Nitrogen Reflow Soldering: No

BOARD CHARACTERISTICS

Copper Layer Count:	8	Board Thickness:	1.6180 mm
Board overall dimensions:	130.0000 mm x 170.0000 mm		
Min track/spacing:	0.1000 mm / 0.1000 mm	Min hole diameter:	0.1500 mm
Copper Finish:	ENIG	Impedance Control:	Yes
Castellated pads:	No	Plated Board Edge:	No
Edge card connectors:	No		

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Item Number: BR-103907		Tolerances Specified per Decimal Place Unless Otherwise Specified	Linear:	Angular:
Description: Sonar Development Board			X. ± 0.2 X.X ± 0.1 X.XX ± 0.05	X. ± 1 X.X ± 0.5 X.XX ± 0.1
Unpop. PCB: BR-103907-001 RevD		PCBA: BR-103907-002 RevD		
Size: USLedge	Date:			Scale: 1:1
KiCad E.D.A. 10.0.6	Sheet:			Id: 6/6